1

Introduction to Mathematics

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Notation

$\mathcal{M}_+(X, \mathcal{A})$	finite measures on $(X, \mathcal{A})$
v	velocity

Preface

cc-by-nc-sa

Part I

Logic and Set Theory

0.1 To Do

- need to merge the set theory notes with this intro to math. It might as well cover all the stuff related to set theory, just with more prose.

Chapter 1

Introduction to Logic

1.0.1 Quantification

- discuss propositional logic and how it can represent statements and how to model it with computer say. Talk about conjunction (and), disjunction (or), implication (if,then), exlusive disjunction (either or) (should discuss the controversy about 'either or'. Just note that human language is often imprecise and establish a convention that 'or' is to be taken in the sense of a choice of soup or salad at a buffet while 'either or' is to be taken in the sense of a choice of soup or salad at a restaurant).

- Motivate quantification and type theory as a way to address the need for a conjucntion/disjunction for infintely many statements while propositional logic only accomodates conjunction of finitely many statements. For example we would like to be able to represent/reason with statements like "there is a positive integer that is larger than 2". We can intuit this in a few ways:

  - One way is with infinitely many disjuctions: "$1 > 2$ or $2 > 2$ or $3 > 2$ or ...". Since this disjunction does not exist in propositional logic, we can just define an extension of propositional logic in which the infinte disjunction exists and is a well-defined operation. This is analogous to how the positive real numbers $[0, \infty)$ does not have a natural infinte summation operation. We resolve both of these issues by extending the objects under consideration, statements ϕ, (elements of $[0, \infty)$ in the analogy), to a larger collection of objects, indexed statements $\phi(x)$, such as "$x > 2$", where we indentify statements from the original collection such as "5 is odd" with the indexed family $\phi(5) :=$ "5 is odd" and $\phi(x) := \perp$ for $x \neq 5$.

    as statements in the original collection of sentences $((a_n)_{n\in\mathbb{N}} \in [0, \infty]^{\mathbb{N}}$, i.e. sequences of nonegative real numbers in the analogy where we identify $a \in [0, \infty)$ with $(a_n)_{n\in\mathbb{N}}$ defined by $a_1 := a$ and $a_n := 0$), and statements $\phi(a)$ indexed by some variable a

    (i.e. we just pretend it exists and throw it in), much like how there is no infinte summation of real numbers, but you can define an infinte summation operation as a limit of partial sums which satisfies the properties of what we usually think an 'addition' operation ought to satisfy.

    Specifically, we say "for each a, if a is an integer, then $a \cdot 0 = 0$". Here the variable a is playing the role of the index of summation $\prod_a x_a$

  - Annother is as an implication: "if a is an integer, then $a \cdot 0 = 0$".

- Motivate type theory as the need for a conjucntion/disjunction for infintely many statements and propositional logic only accomodates finitely many statements.

- Discuss quantification from a set theory perspective where everything is a set and we quantify over sets. For example, we would represent " 'is even', we would write '$\forall x, x$"

- discuss phrasing $\forall$ and $\exists$ in terms of a hypothetical sequence of transactions between two people, so that $f : \mathbb{R} \to \mathbb{R}$ is continuous at $x \in \mathbb{R}$ if $\forall \epsilon > 0, \exists \delta > 0, \forall y \in \mathbb{R}, |x - y| < \delta \implies |f(x) - f(y)| < \epsilon$ can be rephrased as "If you give me an $\epsilon > 0$ (doesn't matter which one you give me), I can then give you a $\delta > 0$ (which may depend of ϵ) such that (i.e. that satisfies/has the property that) if you give me a $y \in \mathbb{R}$ (doesn't matter which one), we have that (I can show you that) if $|x - y| < \delta$, then $|f(x) - f(y)| < \epsilon$."

- discuss the connection between restricted quantification and unrestricted quantification with implication. For example, "$\forall x \in \mathbb{R}$, $x^2 > 0$" can be rephrased as "$\forall x, x \in \mathbb{R} \implies x^2 > 0$". Point out that some sources claim that restricted quantification is

equivalent to implication. For example "$\forall x \in \mathbb{R}$, x" are equivalent and why we use quantification in the first place. For example The reason we

Chapter 2

Introduction to Set Theory

2.1 Sets

In this section, we provide an exposure to a basic framework, called set theory. This framework provides a unifying language which allows us to discuss many objects of interest in mathematics by providing the ability to

1. more precisely define objects of interest,
2. more precisely formulate and prove statements about those objects
3. perform (1) and (2) in a way which is aligned with our intuition.

At an intuitive level, the main idea of a **set** is that of a collection of distinct objects, which we refer to as **elements** or **members**. When denoting a set, it is customary to list the elements of the set, separated by a comma and inside of a pair of curly brackets. Some examples of sets include the following:

- the set of letters of the English alphabet: $\{a, b, \ldots, z\}$, a set with 26 elements
- the set of inhabitants of a specific household: $\{$"Marielle", "Fred", "Dexter"$\}$, a set with 3 elements
- the set of nonnegative even integers: $\{0, 2, \ldots\}$, a set with infintely many elements

Note 2.1.0.1. Since the elements of a set must be distinct, we would consider, for instance, the sets $\{a, b, c\}$ and $\{a, a, b, c\}$ to represent the same set.

When expressing that a given object is a member of set, it is customary to use the **membership symbol**, denoted $\in$. Some examples include the following:

- $2 \in \{1, 2, 3, 4\}$
- "Earth" $\in \{$"Mercury", "Venus", "Earth", "Mars"$\}$.

Note 2.1.0.2. We note that this introduction to set theory is not formal, but intuitve. As such, we make no attempt to precisely define the notion of a set or the notion of membership to a set.

At this point in our discussion, it might seem that any advantage conferred by the language of sets in studying mathematics remains to be seen. Indeed, more time and experience may be required to fully appreciate the power of this framework, but for now we summarize why set theory is relevant in mathematics with the following mantra:

In mathematics, everything is a set.

There is a similar perspective adopted in the context of Linux operating systems where every object is a file or in the context of object oriented programming where every object is a class.

If one already has familiarity with various objects in mathematics such as various numbers, functions, and shapes, they might wonder how these objects could be considered as a set, i.e. a list of elements. For example, which elements $a_1, a_2, \ldots$, which are

themselves sets (because everything is a set), would allow one to express the integer 2 as the set $\{a_1, a_2, \ldots\}$? At first glance, we might attempt to represent 2 with an arbitrary set containing two elements, say {"cow", "chicken"}. The intuition here might be that the number of elements reflects our notion of the "quantity" or "magnitude" we associate to the number 2. However, upon further reflection, such a set might not satisfy other notions we associate to the number 2.

For example we typically think of the number 2 as having a unique "ranking" or "place" in the number line. That is, the number 2 is the integer that preceeds the number 3. So we would need to not only define a set representing the number 3, but also a set representing the notion of subsequence that would align with our intuition that 3 is subsequent to 2.

In a different vein, one might also wish for our representation of the number 2 to agree with the idea that the sum of 2 and 4 is 6. The issue here is how to describe the notion of integer addition. After some thought, one might realize that the idea of adding two integers is basically that of a function, i.e. an object which eats two integer inputs and spits out an integer output.

It might seem tricky to be able to represent all the nonnegative integers, the notion of integer subsequence and the notion of integer addition by sets in a way that aligns with our intution of those objects, relations and operations. With our limited toolset, this would indeed be quite challenging, but after a few more ideas, this will become a relatively simple goal. For now, we focus on providing the necessary calculus of sets to be able to do so.

The first two relations between sets that we define pertain to when one set is a subset of another and when one set is equal to another. To motivate this, we observe that each element of $\{a, b\}$ is also an element of $\{a, b, c\}$. Our intuition dictates that the former is in some sense "less than or equal to" the latter. However, the same could not be said of the converse. Therefore, the two sets should not be considered equal.

Definition 2.1.0.3. Let A and B be sets. Then

- A is said to be a **subset of** B if for each set x, $x \in A$ implies that $x \in B$.
  (A is a subset of $B \iff \forall x(x \in A \implies x \in B)$)
- A is said to be **equal to** B if $A \subset B$ and $B \subset A$.
- A is said to be **empty** if for each set x, $x \notin A$.
  (A is empty $\iff \forall x(\neg(x \in A))$)

Note 2.1.0.4.

- When defining a term, it is conventional to use an "if", but really the relationship is an "iff".
- For convenience, we will typically write $x \notin A$ in place of $\neg(x \in A)$.

The following exercise shows us that there is at most one set that is empty.

Exercise 2.1.0.5. Let A, B be sets. If A is empty and B is empty, then $A = B$.

Proof. Suppose that A is empty and B is empty. Let x be a set.

- Since A is empty, $x \notin A$. Therefore $x \in A$ implies that $x \in B$. Since x is an arbitrary set, we have that for each set x, $x \in A$ implies that $x \in B$. Hence $A \subset B$.
- Similarly $B \subset A$

Since $A \subset B$ and $B \subset A$, we have that $A = B$. □

Even though there can be at most one set that is empty, we are not guaranteed the existence of such a set. Since its existence is quite useful and intuitively quite reasonable, we will assume such a set does in fact exist.

Axiom 2.1.0.6. There exists a set $\varnothing$, referred to as the **empty set**, such that for each set x, $x \notin \varnothing$.

Exercise 2.1.0.7. Let A be a set. Then $\varnothing \subset A$.

Proof. Let x be a set. Then $x \notin \varnothing$. Hence $x \in \varnothing$ implies that $x \in A$. Since x is an arbitrary set, we have that for each set x, $x \in \varnothing$ implies that $x \in A$. Thus $\varnothing \subset A$. □

give intuition for union, intersection, complement, show we can define union in terms of intersection and complement, call back to demorgans laws for logic

Definition 2.1.0.8. Let A, B be sets. We define intersection, union, empty set

discuss and introduce/make pictures of venn diagrams

discuss how definitions by convention use an if, but it should be treated as an iff. The if makes sense because if we say that a dog is "blarge" if it is black and weighs more than 15 kg, if we had put an iff in this definition, then the we would be saying that if a dog is blarge, then it is black and weighs more than 15 kgs, which wouldn really make sense in a definition because we have not defined previously and do not know what it is blarge, so it is a language thing. but really we are defining a set. $x \in \{\text{blarge dogs}\} \iff x \in \{\text{black dogs}\}\text{and}\{\text{dogs above 15 kg}\}$ i.e. $\{\text{blarge dogs}\} = \{\text{black dogs}\} \cap \{\text{dogs above 15 kg}\}$.

discuss how we can define natural numbers

discuss how we define $(a, b) = \{\{a\}, \{a, b\}\}$ or similar

Definition 2.1.0.9. products $A \times B$

Exercise 2.1.0.10. content...

Exercise 2.1.0.11.

2.2 Relations

2.3 Functions

2.3.1 Introduction

Definition 2.3.1.1. define function, Let X, Y be sets.

- Let $f \subset \mathcal{R}(A, B)$. Then f is said to be a **function from** X **to** Y, denoted $f : X \to Y$, if
  1. for each $x \in X$, there exists $y \in Y$ such that $(x, y) \in f$,
  2. for each $x_1, x_2 \in X$, if $x_1 = x_2$, then $f(x_1) = f(x_2)$.
- We define $Y^X := \{f : f : X \to Y\}$.

later in section on products, show that $Y^X = \prod_{x \in X} A$

Exercise 2.3.1.2. Let X be a set. If $X \neq \varnothing$, then $\varnothing^X = \varnothing$.

Proof. For the sake of contradiction, suppose that $X \neq \varnothing$ and $\varnothing^X \neq \varnothing$. Then there exists $f : X \to \varnothing$. Since $X \neq \varnothing$, there exists $x \in X$. Since f is a function, there exists $y \in \varnothing$ such that $f(x) = y$. This is a contradiction. Hence $\varnothing^X = \varnothing$. □

Definition 2.3.1.3. Let X, Y be sets and $f : X \to Y$. Then f is said to be

- **injective** or an **injection** if for each $x_1, x_2 \in X$, $f(x_1) = f(x_2)$ implies that $x_1 = x_2$.
- **surjective** or a **surjection** if for each $y \in Y$, there exists $x \in X$ such that $f(x) = y$
- **bijective** a **bijection** if f is injective and f is surjective.

Exercise 2.3.1.4. Let X, Y be sets and $f : X \to Y$. Then

1. f is injective iff for each $y \in f(X)$, there exists $x \in X$ such that $f^{-1}(\{y\}) = \{x\}$,
2. f is surjective iff $f(X) = Y$.

Proof.

1.
   - ($\Longrightarrow$):
     Supose that f is injective. Let $y \in f(X)$. Then there exists $x \in X$ such that $f(x) = y$.
     - Since
       $$\begin{aligned} f(x) &= y \\ &\in \{y\}, \end{aligned}$$
       we have that $x \in f^{-1}(\{y\})$. Therefore $\{x\} \subset f^{-1}(\{y\})$.
     - Let $x' \in f^{-1}(\{y\})$. By definition (ref def), $f(x') \in \{y\}$. Exercise **??** (ref ex) implies that $f(x') = y$. Therefore
       $$\begin{aligned} f(x') &= y \\ &= f(x). \end{aligned}$$
       Since f is injective, $x' = x$. Hence $x' \in \{x\}$. Since $x' \in f^{-1}(\{y\})$ is arbitrary, we have that for each $x' \in f^{-1}(\{y\})$, $x' \in \{x\}$. Therefore $f^{-1}(\{y\}) \subset \{x\}$.

     Since $\{x\} \subset f^{-1}(\{y\})$ and $f^{-1}(\{y\}) \subset \{x\}$, we have that $f^{-1}(\{y\}) = \{x\}$. Since $y \in f(X)$ is arbitrary, we have that for each $y \in f(X)$, there exists $x \in X$ such that $f^{-1}(\{y\}) = \{x\}$.
   - ($\Longleftarrow$):
     Suppose that for each $y \in f(X)$, there exists $x \in X$ such that $f^{-1}(\{y\}) = \{x\}$. Let $x_1, x_2 \in X$. Suppose that $f(x_1) = f(x_2)$. Define $y \in f(X)$ by $y := f(x_1)$. By assumption, there exists $x \in X$ such that $f^{-1}(\{y\}) = \{x\}$. Since
     $$\begin{aligned} f(x_1) &= y \\ &\in \{y\}, \end{aligned}$$

we have that $x_1 \in f^{-1}(\{y\})$. Therefore

$$\begin{aligned} x_1 &\in f^{-1}(\{y\}) \\ &= \{x\}. \end{aligned}$$

Similarly, $x_2 \in \{x\}$. Exercise **??** (ref ex) then implies that

$$\begin{aligned} x_1 &= x \\ &= x_1. \end{aligned}$$

Since $x_1, x_2 \in X$ with $f(x_1) = f(x_2)$ are arbitrary, we have that for each $x_1, x_2 \in X$, $f(x_1) = f(x_2)$ implies that $x_1 = x_2$. Thus f is injective.

2.
   - ($\Longrightarrow$):
     Supose that f is surjective.
     - Definition **??** (ref def) implies that $f(X) \subset Y$.
     - Let $y \in Y$. Since f is surjective, there exists $x \in X$ such that $f(x) = y$. Then

       $$\begin{aligned} y &= f(x) \\ &\in f(X). \end{aligned}$$

       Since $y \in Y$ is arbitrary, we have that for each $y \in Y$, $y \in f(X)$. Hence $Y \subset f(X)$.

     Since $f(X) \subset Y$ and $Y \subset f(X)$, we have that $f(X) = Y$.
   - ($\Longleftarrow$):
     Suppose that $f(X) = Y$. Let $y \in Y$. Since $f(X) = Y$, we have that $y \in f(X)$. Definition **??** (ref def) implies that there exists $x \in X$ such that $f(x) = y$. Since $y \in Y$ is arbitrary, we have that for each $y \in Y$, there exists $x \in X$ such that $f(x) = y$. Hence f is surjective.

□

2.3.2 Axiom of Choice

Definition 2.3.2.1. Let A, B be sets, $F : A \to \mathcal{P}(B)$ and $f : A \to B$. Then f is said to be a **selection of** F if for each $a \in A$, $f(a) \in F(a)$.

Axiom 2.3.2.2. Axiom of Choice:
Let A, B be sets, $F : A \to \mathcal{P}(B)$. If for each $a \in A$, $F(a) \neq \varnothing$, then there exists $f : A \to B$ such that f is a selection of F.

2.3.3 Composition

Definition 2.3.3.1. Let X, Y, Z be sets, $f : X \to Y$ and $g : Y \to Z$. We define the **composition of** g **with** f, denoted $g \circ f : X \to Y$, by

$$g \circ f(x) := g(f(x)) \quad (x \in X)$$

Exercise 2.3.3.2. Let X, Y, Z be sets, $f : X \to Y$ and $g : Y \to Z$. Then $g \circ f : X \to Y$.

Proof.

1. Let $x \in X$. Since $f : X \to Y$ and $x \in X$, $f(x) \in Y$. Define $y \in Y$ by $y := f(x)$. Since $g : Y \to Z$ and $y \in Y$, $g(y) \in Z$. Therefore

   $$\begin{aligned} g \circ f(x) &= g(f(x)) \\ &= g(y) \\ &\in Z. \end{aligned}$$

   Since $x \in X$ is arbitrary, we have that for each $x \in X$, $g \circ f(x) \in Z$.

2. Let $x_1, x_2 \in X$. Suppose that $x_1 = x_2$. Define $y_1, y_2 \in Y$ by $y_1 := f(x_1)$ and $y_2 := f(x_2)$. Since $f : X \to Y$ and $x_1 = x_2$, we have that

$$\begin{aligned} y_1 &= f(x_1) \\ &= f(x_2) \\ &= y_2. \end{aligned}$$

Since $g : Y \to Z$ and $y_1 = y_2$, we have that

$$\begin{aligned} g \circ f(x_1) &= g(f(x_1)) \\ &= g(y_1) \\ &= g(y_2) \\ &= g(f(x_2)) \\ &= g \circ f(x_2). \end{aligned}$$

Since $x_1, x_2 \in X$ with $x_1 = x_2$ are arbitrary, we have that for each $x_1, x_2 \in X$, $x_1 = x_2$ implies that $g \circ f(x_1) = g \circ f(x_2)$.

Thus $g \circ f : X \to Y$. □

Exercise 2.3.3.3. Let X, Y, Z, W be sets, $f : X \to Y$, $g : Y \to Z$ and $h : Z \to W$. Then $(h \circ g) \circ f = h \circ (g \circ f)$.

Proof. For each $x \in X$,

$$\begin{aligned} (h \circ g) \circ f(x) &= (h \circ g)(f(x)) \\ &= h(g(f(x))) \\ &= h(g \circ f(x)) \\ &= h \circ (g \circ f)(x). \end{aligned}$$

Thus $(h \circ g) \circ f = h \circ (g \circ f)$. □

Definition 2.3.3.4. Let X be a set. We define the **identity function on** X, denoted $\mathrm{id}_X : X \to X$ by $\mathrm{id}_X(x) := x \quad (x \in X)$.

Exercise 2.3.3.5. Let X be a set. Then $\mathrm{id}_X : X \to X$.

Proof.

1. By definition, for each $x \in X$,

$$\begin{aligned} \mathrm{id}_X(x) &= x \\ &\in X. \end{aligned}$$

2. By definition, for each $x_1, x_2 \in X$, if $x_2 = x_2$, then

$$\begin{aligned} \mathrm{id}_X(x_1) &= x_1 \\ &= x_2 \\ &= \mathrm{id}_X(x_2). \end{aligned}$$

Thus $\mathrm{id}_X : X \to X$. □

Exercise 2.3.3.6. Let X, Y be sets and $f : X \to Y$. Then

1. $f \circ \mathrm{id}_X = f$,
2. $\mathrm{id}_Y \circ f = f$.

Proof.

1. For each $x \in X$,

$$\begin{aligned} f \circ \mathrm{id}_X(x) &= f(\mathrm{id}_X(x)) \\ &= f(x). \end{aligned}$$

   Thus $f \circ \mathrm{id}_X = f$.

2. For each $y \in Y$,

$$\begin{aligned} \mathrm{id}_Y \circ f(x) &= \mathrm{id}_Y(f(x)) \\ &= f(x). \end{aligned}$$

   Thus $\mathrm{id}_Y \circ f = f$.

□

Definition 2.3.3.7. Let X, Y be sets and $f : X \to Y$.

- Let $g : Y \to X$. Then g is said to be a(n)
  - **left inverse of** f if $g \circ f = \mathrm{id}_X$,
  - **right inverse of** f if $f \circ g = \mathrm{id}_Y$,
  - **inverse of** f if g is a left inverse of f and g is a right inverse of f.
- Then f is said to
  - have a(n)
    - **left inverse** if there exists $g : Y \to X$ such that g is a left inverse of f,
    - **right inverse** if there exists $g : Y \to X$ such that g is a right inverse of f.
    - **inverse** if there exists $g : Y \to X$ such that g is an inverse of f,
  - be
    - **left invertible** if f has a left inverse,
    - **right invertible** if f has a right inverse,
    - **invertible** if f has an inverse.

Exercise 2.3.3.8. make ex about nonunique left and right inverses

Proof. finish!!! □

Exercise 2.3.3.9. Let X, Y be sets, $f : X \to Y$, $g, g' : Y \to X$. If g is an inverse of f and g' is an inverse of f, then $g = g'$.

Proof. Suppose that g is an inverse of f and g' is an inverse of f. Then

$$\begin{aligned} g' &= g' \circ \mathrm{id}_Y \\ &= g' \circ (f \circ g) \\ &= (g' \circ f) \circ g \\ &= \mathrm{id}_X \circ g \\ &= g. \end{aligned}$$

□

Definition 2.3.3.10. Let X, Y be sets and $f : X \to Y$. Suppose that f is invertible. We define the **inverse of** f, denoted $f^{-1} : Y \to X$, to be the unique $g : Y \to X$ such that g is an inverse of f.

Composition and X-jectivity

Exercise 2.3.3.11. Let X be a set. Then

1. id_X is injective,
2. id_X is surjective,
3. id_X is bijective.

Proof.

1. For each $x_1, x_2 \in X$, $\mathrm{id}_X(x_1) = \mathrm{id}_X(x_2)$ implies that

$$\begin{aligned} x_1 &= \mathrm{id}_X(x_1) \\ &= \mathrm{id}_X(x_2) \\ &= x_2. \end{aligned}$$

   Thus id_X is injective.

2. Let $y \in X$. Define $x \in X$ by $x := y$. Then

$$\begin{aligned} \mathrm{id}_X(x) &= x \\ &= y. \end{aligned}$$

   Since $y \in X$ is arbitrary, we have that for each $y \in X$, there exists $x \in X$ such that $\mathrm{id}_X(x) = y$. Hence id_X is surjective.

3. Since id_X is injective and id_X is surjective, we have that id_X is bijective.

□

Exercise 2.3.3.12. Let X, Y, Z be sets, $f : X \to Y$ and $g : Y \to Z$.

1. If g is injective and f is injective, then $g \circ f$ is injective.
2. If g is surjective and f is surjective, then $g \circ f$ is surjective.
3. If g is bijective and f is bijective, then $g \circ f$ is bijective.

Proof.

1. Suppose that g is injective and f is injective. Let $x_1, x_2 \in X$. Suppose that $g \circ f(x_1) = g \circ f(x_2)$. Define $y_1, y_2 \in Y$ by $y_1 := f(x_1)$ and $y_2 := f(x_2)$. Then

$$\begin{aligned} g(y_1) &= g(f(x_1)) \\ &= g \circ f(x_1) \\ &= g \circ f(x_2) \\ &= g(f(x_2)) \\ &= g(y_2). \end{aligned}$$

   Since g is injective, we have that $y_1 = y_2$. Therefore

$$\begin{aligned} f(x_1) &= y_1 \\ &= y_2 \\ &= f(x_2). \end{aligned}$$

   Since f is injective, we have that $x_1 = x_2$. Since $x_1, x_2 \in X$ with $g \circ f(x_1) = g \circ f(x_2)$ are arbitrary, we have that for each $x_1, x_2 \in X$, $g \circ f(x_1) = g \circ f(x_2)$ implies that $x_1 = x_2$. Thus $g \circ f$ is injective.

2. Suppose that g is surjective and f is surjective. Let $z \in Z$. Since g is surjective, there exists $y \in Y$ such that $g(y) = z$. Since f is surjective, there exists $x \in X$ such that $f(x) = y$. Then

$$\begin{aligned} g \circ f(x) &= g(f(x)) \\ &= g(y) \\ &= z. \end{aligned}$$

Since $z \in Z$ is arbitrary, we have that for each $z \in Z$, there exists $x \in X$ such that $g \circ f(x) = z$. Hence $g \circ f$ is surjective.

3. Suppose that g is bijective and f is bijective. Then g is injective, g is surjective, f is injective and f is surjective. the prev parts (ref parts) implies that $g \circ f$ is injective and $g \circ f$ is surjective. Thus $g \circ f$ is bijective.

□

Exercise 2.3.3.13. Let X, Y, Z be sets, $f : X \to Y$ and $g : Y \to Z$. Then

1. g is injective implies that $g \circ f$ is injective iff f is injective.
2. f is surjective implies that $g \circ f$ is surjective iff g is surjective.

Proof.

1. Suppose that g is injective.
   - ($\Longrightarrow$):
     Suppose that $g \circ f$ is injective. Let $x_1, x_2 \in X$. Suppose that $f(x_1) = f(x_2)$. Define $y_1, y_2 \in Y$ by $y_1 := f(x_1)$ and $y_2 := f(x_2)$. By assumption,

$$\begin{aligned} y_1 &= f(x_1) \\ &= f(x_2) \\ &= y_2. \end{aligned}$$

     Since $g : Y \to Z$, we have that $g(y_1) = g(y_2)$. Therefore

$$\begin{aligned} g \circ f(x_1) &= g(f(x_1)) \\ &= g(y_1) \\ &= g(y_2) \\ &= g(f(x_2)) \\ &= g \circ f(x_2). \end{aligned}$$

     Since $g \circ f$ is injective, we have that $x_1 = x_2$. Since $x_1, x_2 \in X$ with $f(x_1) = f(x_2)$ are arbitrary, we have that for each $x_1, x_2 \in X$, $f(x_1) = f(x_2)$ implies that $x_1 = x_2$. Thus f is injective.
   - ($\Longleftarrow$):
     Since f is injective, Exercise **??** (ref prev ex) implies that if f is injective, then $g \circ f$ is injective.
2. Suppose that f is surjective.
   - ($\Longrightarrow$):
     Suppose that $g \circ f$ is surjective. Let $z \in Z$. Since $g \circ f$ is surjective, there exists $x \in X$ such that $g \circ f(x) = z$. Define $y \in Y$ by $y := f(x)$. Then

$$\begin{aligned} g(y) &= g(f(x)) \\ &= g \circ f(x) \\ &= z. \end{aligned}$$

     Since $z \in Z$ is arbitrary, we have that for each $z \in Z$, there exists $y \in Y$ such that $g(y) = z$. Thus g is surjective.

- ($\Longleftarrow$):
  Since f is surjective, Exercise **??** (ref prev ex) implies that if g is surjective, then $g \circ f$ is surjective.

□

Exercise 2.3.3.14. Let X, Y be sets and $f : X \to Y$. Then

1. f is left invertible iff f is injective.
2. f is right invertible iff f is surective.
3. f is invertible iff f is a bijection.

Proof.

1. - ($\Longrightarrow$):
     Supose that f is left invertible. Then f has a left inverse. Thus there exists $g : Y \to X$ such that $g \circ f = \mathrm{id}_X$. Therefore for each $x_1, x_2 \in X$, $f(x_1) = f(x_2)$ implies that

     $$\begin{aligned} x_1 &= \mathrm{id}_X(x_1) \\ &= g \circ f(x_1) \\ &= g(f(x_1)) \\ &= g(f(x_2)) \\ &= g \circ f(x_2) \\ &= \mathrm{id}_X(x_2) \\ x_2. \end{aligned}$$

     Hence f is injective.
   - ($\Longleftarrow$):
     - Suppose that f is injective. Exercise **??** (ref ex) implies that for each $y \in f(X)$, there exists $x_y \in X$ such that $f^{-1}(\{y\}) = \{x_y\}$. Define $F : Y \to \mathcal{P}(X)$ by $F(y) := f^{-1}(\{y\})$. Since for each $y \in Y$, $F(y) \neq \varnothing$, the axiom of choice (ref axiom) implies that there exists $g : Y \to X$ such that for each $y \in Y$, $g(y) \in F(y)$. Exercise **??** (ref ex) implies that for each $y \in Y$, $g(y) = x_y$.
     - Let $x \in X$. Define $y \in Y$ by $y := f(x)$. By definition of g,

       $$\begin{aligned} x_y &= g(y) \\ &\in f^{-1}(\{y\}). \end{aligned}$$

       Thus

       $$\begin{aligned} f(x_y) &= y \\ &= f(x). \end{aligned}$$

       Since f is injective, we have that

       $$\begin{aligned} x_{f(x)} &= x_y \\ &= x. \end{aligned}$$

       Since $x \in X$ is arbitrary, we have that for each $x \in X$, $x_{f(x)} = x$.
     - Then for each $x \in X$,

       $$\begin{aligned} g \circ f(x) &= g(f(x)) \\ &= x_{f(x)} \\ &= x \\ &= \mathrm{id}_X(x). \end{aligned}$$

       Hence $g \circ f = \mathrm{id}_X$. Thus f has a left inverse and therefore f is left invertible.

2. • ($\Longrightarrow$):
   Supose that f is right invertible. Then f has a right inverse. Thus there exists $g : Y \to X$ such that $f \circ g = \mathrm{id}_Y$. Hence f is surjective.
   • ($\Longleftarrow$):
   – Suppose that f is surjective.
3.

□

Exercise 2.3.3.15. Let X, Y be sets and $f : X \to Y$. Then f is invertible iff f is a bijection.

Proof. maybe make an exercise characterizing injectivity and surjectivity in terms of left and right inverses (basically the proof below) and then just reference that exercise here, need axiom of choice and previous exercise about product of singletons being singleton

- ($\Longrightarrow$):
  Suppose that f is invertible.
  – For each $x_1, x_2 \in X$, $f(x_1) = f(x_2)$ implies that

$$\begin{aligned} x_1 &= \mathrm{id}_X(x_1) \\ &= f^{-1} \circ f(x_1) \\ &= f^{-1}(f(x_1)) \\ &= f^{-1}(f(x_2)) \\ &= f^{-1} \circ f(x_2) \\ &= \mathrm{id}_X(x_2) \\ x_2. \end{aligned}$$

  Thus f is injective.
  – Let $y \in Y$. Define $x \in X$ by $x := f^{-1}(y)$. Then

$$\begin{aligned} f(x) &= f(f^{-1}(y)) \\ &= f \circ f^{-1}(y) \\ &= \mathrm{id}_Y(y) \\ &= y. \end{aligned}$$

  Since $y \in Y$ is arbitrary, we have that for each $y \in Y$, there exists $x \in X$ such that $f(x) = y$. Hence f is surjective.
- ($\Longleftarrow$):
  Suppose that f is a bijection. Then f is injective and surjective. Since f is surjective, $f(X) = Y$. Therefore for each $y \in Y$, $f^{-1}(\{y\}) \neq \varnothing$ (define $f^{-1}(A)$ and $f(A)$, show $f(X) = Y$ iff f is surjective and ref exercises). Since f is injective and $f(X) = Y$, for each $y \in f(X)$, there exists $x_y \in X$ such that $f^{-1}(\{y\}) = \{x_y\}$. The axiom of choice (actually an exercise I proved somewhere about products of singleton sets being a singelton set) implies that there exists a unique $g : Y \to X$ such that for each $y \in Y$, $g(y) = x_y$. Then for each $y \in Y$,

$$\begin{aligned} f \circ g(y) &= f(g(y)) \\ &= f(x_y) \\ &= y \\ &= \mathrm{id}_Y(y). \end{aligned}$$

  Thus $f \circ g = \mathrm{id}_Y$. Let $x \in X$. Since f is injective,

$$\begin{aligned} \{x\} &= f^{-1}(\{f(x)\}) \\ &= \{g(f(x))\} \quad \text{(by definition of } g). \end{aligned}$$

Therefore

$$\begin{aligned} g \circ f(x) &= g(f(x)) \\ &= x \\ &= \mathrm{id}_X(x). \end{aligned}$$

Since $x \in X$ is arbitrary, we have that for each $x \in X$, $g \circ f(x) = \mathrm{id}_X(x)$. Hence $g \circ f = \mathrm{id}_X$.

Thus there exists $g : Y \to X$ such that $g \circ f = \mathrm{id}_X$ and $f \circ g = \mathrm{id}_Y$. Hence f is invertible. □

Motivate Schroder-Bernstein theorem with Hilbert's hotel and reference blargoner's youtube video https://www.youtube.com/watch?v=
Write some stuff about hilbert's hotel

Question 2.3.3.16. Suppose we have a hotel with infintely many rooms, each of which is current occupied by a unique guest. Suppose infintely many more prospective guests arrive in search of a room.

Now assume that

- there is a room-assignment procedure which will assign a room to each of the guests (prospective or current) but will not assign more than one guest to the same room,
- there is a occupant-identification procedure which for each room, allows us to identify the current occupant.

We note that the room assingment procedure may not necessarily fill all the rooms and the occupant identification procedure may not necessarily provide the names of all guests (e.g. the prospective guests before they are assigned a room).

Is it possible to assign a room to each guest (current or prospective) using the room-assignment and occupant-identification procedures in such a way so that each of the rooms in the hotel are occupied

draw picture

Solution 2.3.3.17. We will denote the set of guests (current or prospective) by Y and the set of rooms by X. We denote the room-assignment procedure by $g : Y \to X$ and we denote the occupant-identification procedure by $f : X \to Y$.

We may begin by

1. assigning each of the prospective guests to a room using the room-assingment procedure, that is we will send each $y \in f(X)^c$ to $g(y) \in g[f(X)^c]$.
2. Since the rooms in $g[f(X)^c]$ are already currently occupied, for each room in $g[f(X)^c]$, we may identify its occupant using the occupant-identification procedure and temporarily remove them from their room, that is send each $x \in g[f(X)^c]$ to $f(x) \in (f \circ g)[f(X)^c]$.

We may then inductively repeat the previous steps so that after the k-th iteration of the previous steps, we have just assigned each guest $y \in (f \circ g)^{k-1}[f(X)^c]$ to room $g(y) \in g \circ (f \circ g)^{k-1}[f(X)^c]$ and temporarily removed each guest $f(x) \in (f \circ g)^k[f(X)^c]$.

rework intuition, move intuition about size to exercises about injectivity and surjectivity, draw pictures, add picture and intuition about same size-ness for bijections, draw strings, etc

Let X and Y be sets. Suppose that there exists $f : X \to Y$ such that f is injective. Intuitively, we are able to consider Y as being "at least as large as" X because for each $x \in X$, we may tie a string from x to a unique $f(x) \in Y$, so that there are "at least as many" elements of Y as there are elements in X. In this analogy, it is not generally the case that each $y \in Y$ will have a string attached to it unless f is a bijection. However, what if we find ourselves in the situation in which there also exists $g : Y \to X$ such that g is an injection. Then intuitively, X "at least as large as" Y and we would then want it to be the case that X and Y are "the same size". Taking our analogy further, we would like it to be the case that there is a bijection $h : X \to Y$. The previous question about hotels provides the mechanism for showing this to be the case.

Exercise 2.3.3.18. Schroder-Bernstein Theorem:
Let X, Y be sets. If there exist $f : X \to Y$, $g : Y \to X$ such that f and g are injections, then there exists $\phi : X \to Y$ such that ϕ is a bijection.

Proof. Define $A_0 := f(X)^c$ and $A := \bigcup_{k=0} (f \circ g)^{\circ k}(A_0)$. Show $A^c \subset f(X)$ and $f(A^c) = g(A)^c$ and show $g|_A^{g(A)}$ and $(f|_{g(A)^c}^{A^c})^{-1}$ are invertible. Define $h : Y \to X$ by

$$h(y) := \begin{cases} g(y), & y \in A \\ (f|^{f(X)})^{-1}(y), & y \in A^c \end{cases}$$

i.e. $h = g|_A^{g(A)} \oplus (f|_{g(A)^c}^{A^c})^{-1}$
there may be an exercise showing that inverse of coprod is coprod of inverses $\square$

Chapter 3

Cardinality

3.1 Ordinal Numbers

3.2 Cardinal Numbers

Chapter 4

Cardinality